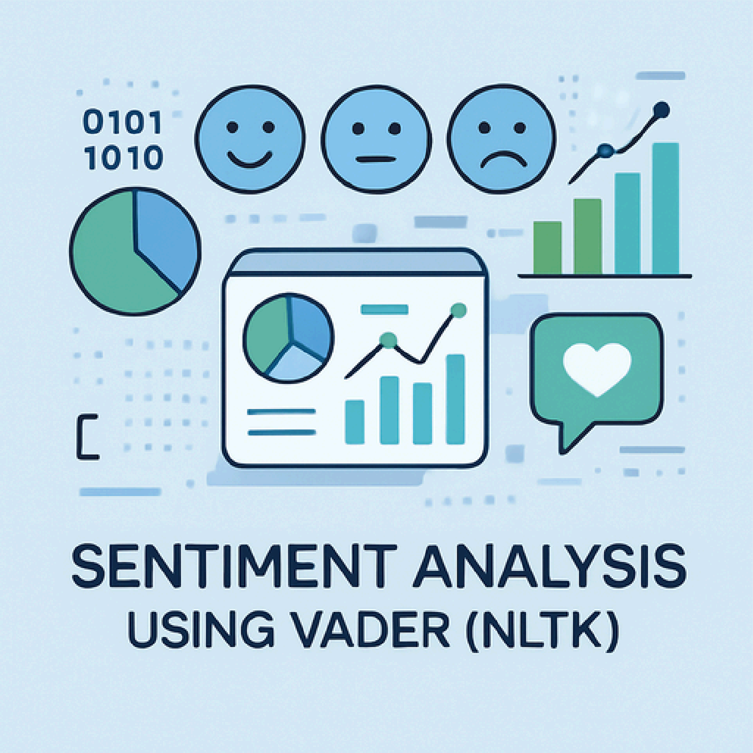
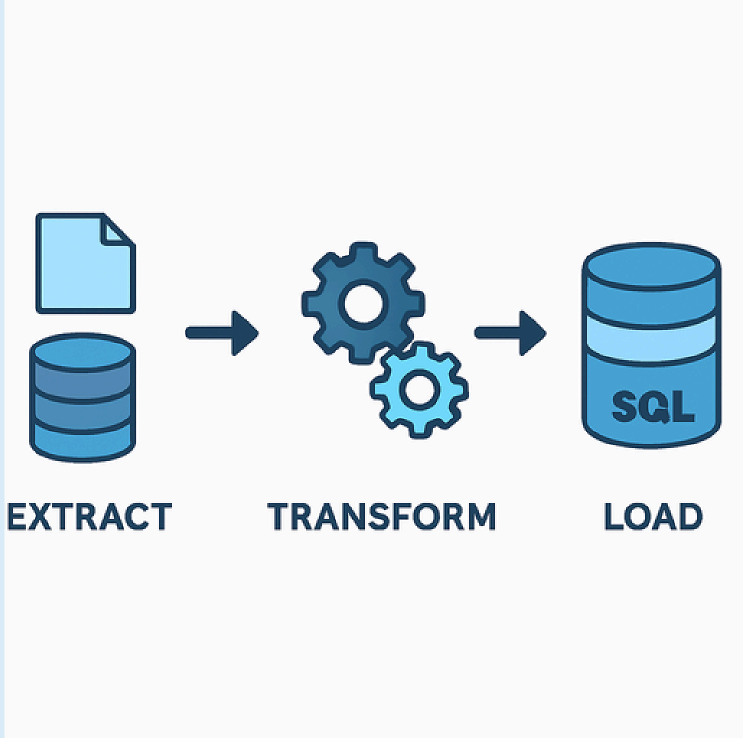
MODERN DATA WAREHOUSE & ANALYTICS PROJECT

MEDALLION ARCHITECTURE | ETL PIPELINES | SENTIMENT ANALYSIS | POWER BI

SQL | POWER BI | VADER (NLTK) | GITHUB LINK

https://github.com/Daven88

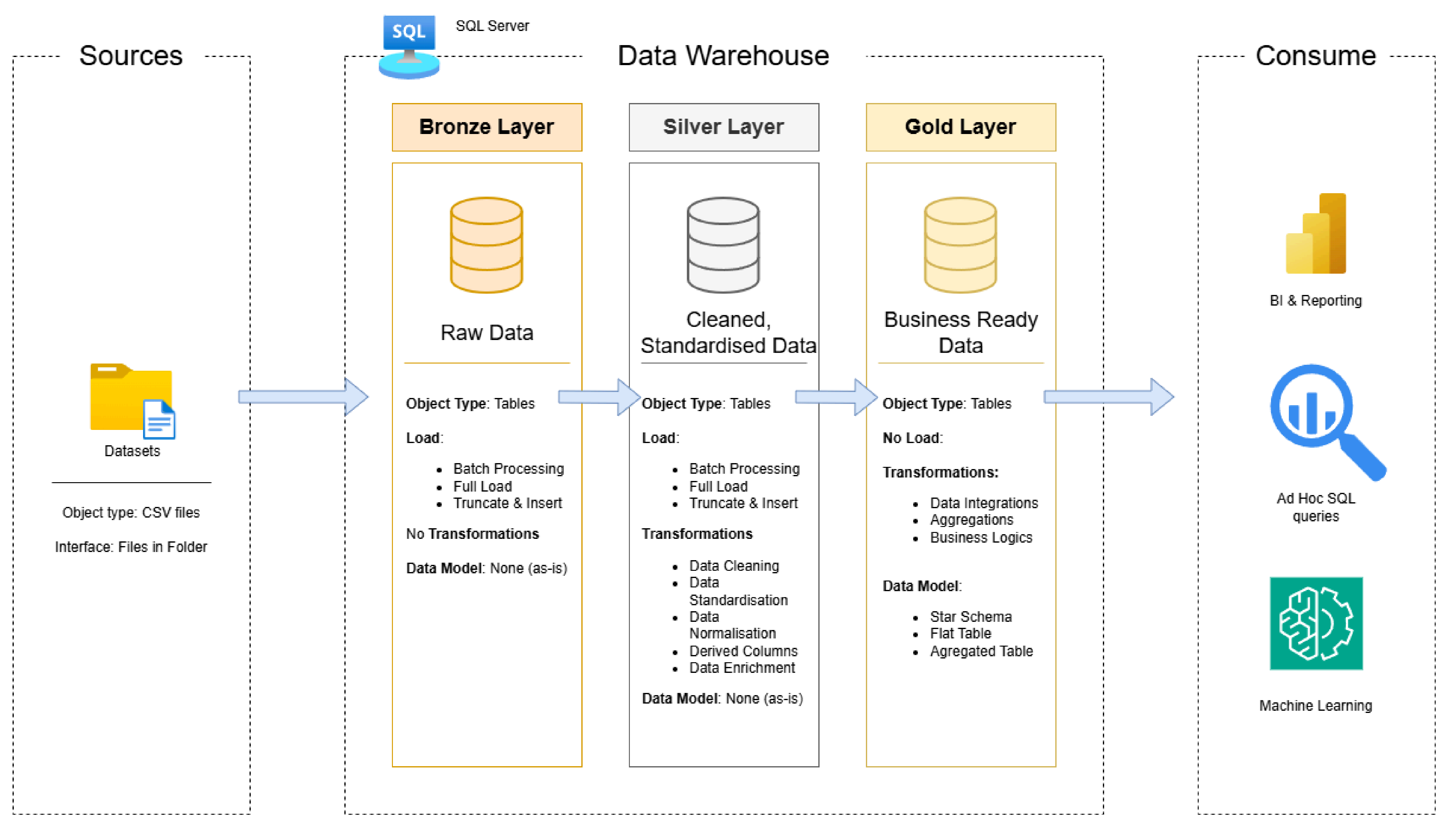




17/02/2025

DATA ARCHITECTURE DESIGN

High Level Architecture

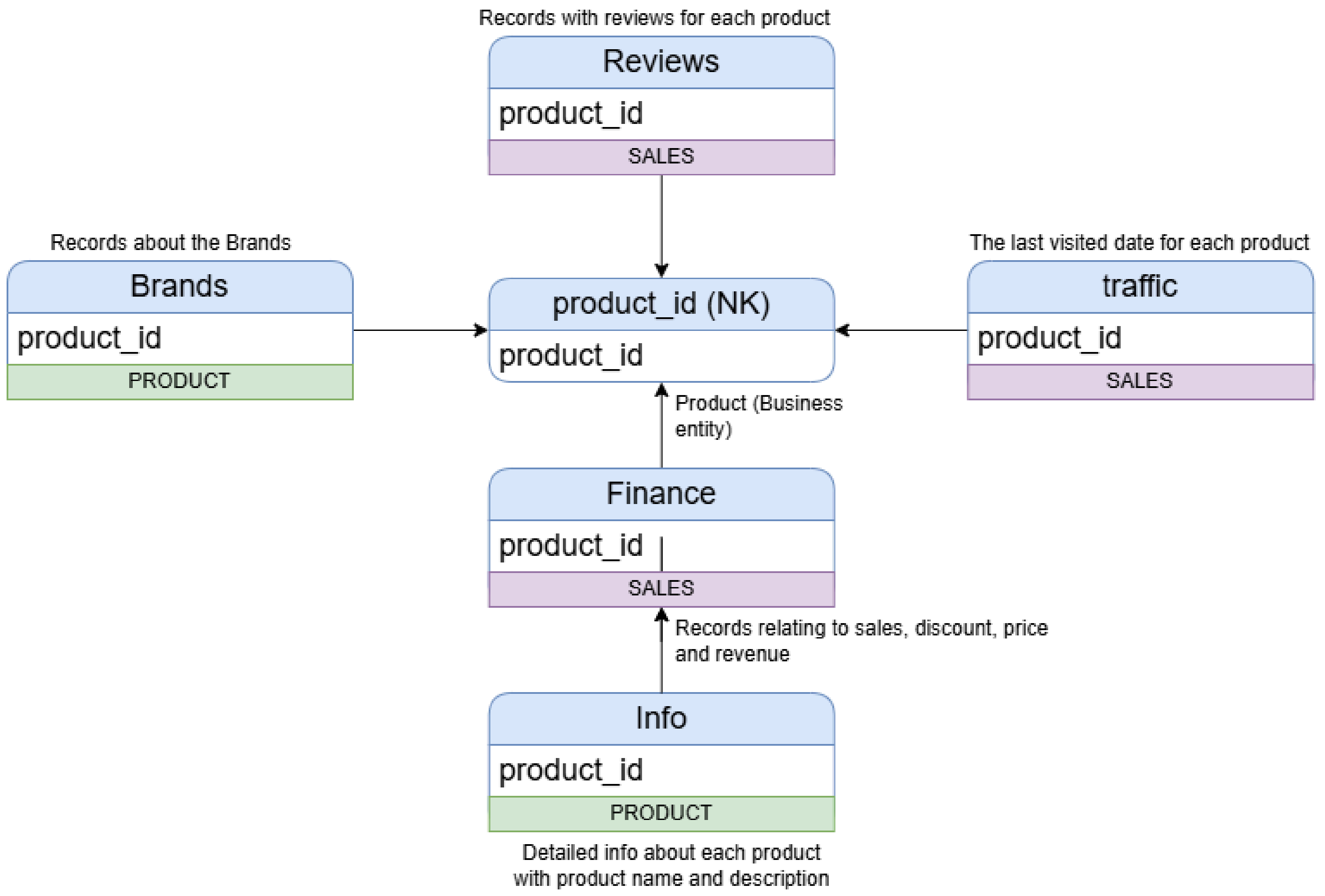


MEDALION ARCHITECTURE

- Bronze – Raw CSV ingestion
- Silver – Data cleaning & transformation
- Gold – Star schema for analytics

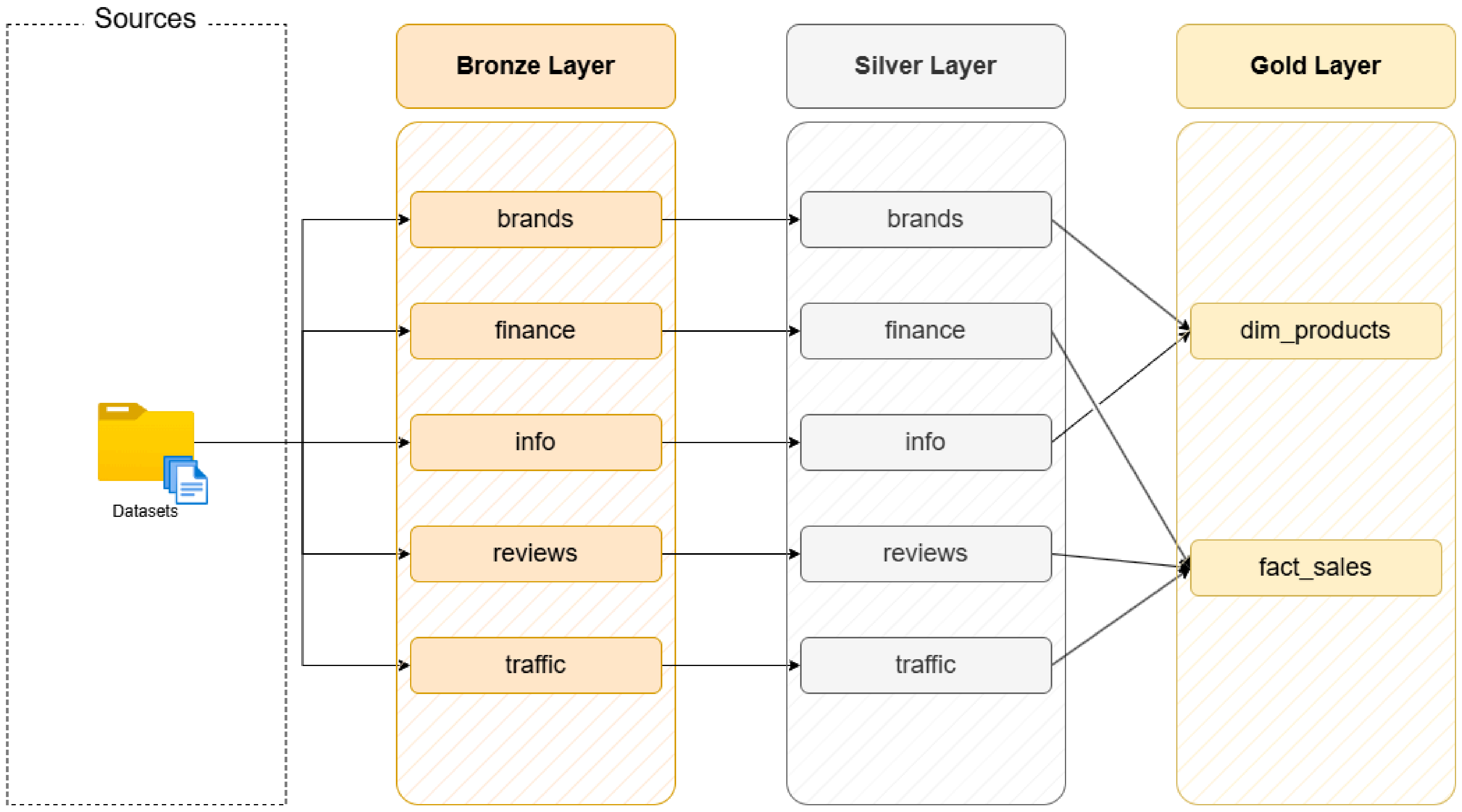
DIMESNIONAL MODELING

Integration Model

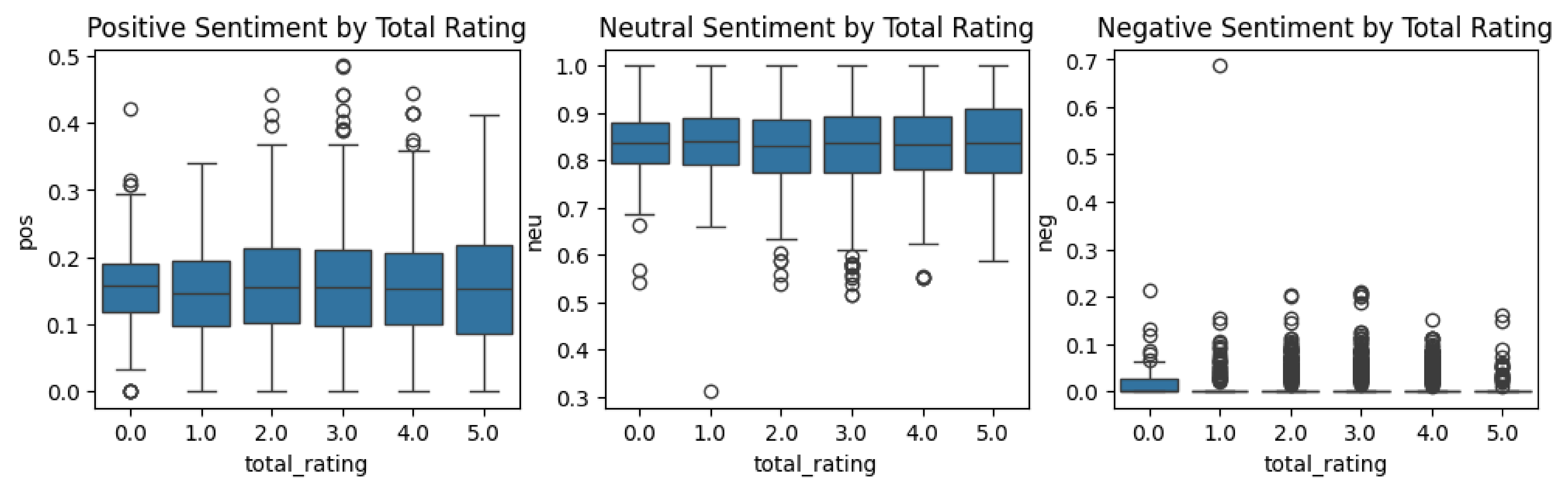


- FACT TABLE: SALES
- DIMENSIONS: PRODUCT
- OPTIMISED FOR ANALYTICAL QUERYING

Data Flow Diagram



- Ingested raw CSV data into Bronze layer (SQL Server)
- Applied cleansing and transformation logic in Silver layer
- Standardised and validated key fields (dates, product IDs))
- Loaded curated data into Gold layer (star schema)
- Enabled downstream analytics and Power BI reporting



- Applied VADER sentiment scoring
- Analysed product reviews & descriptions
- Integrated results into analytics layer

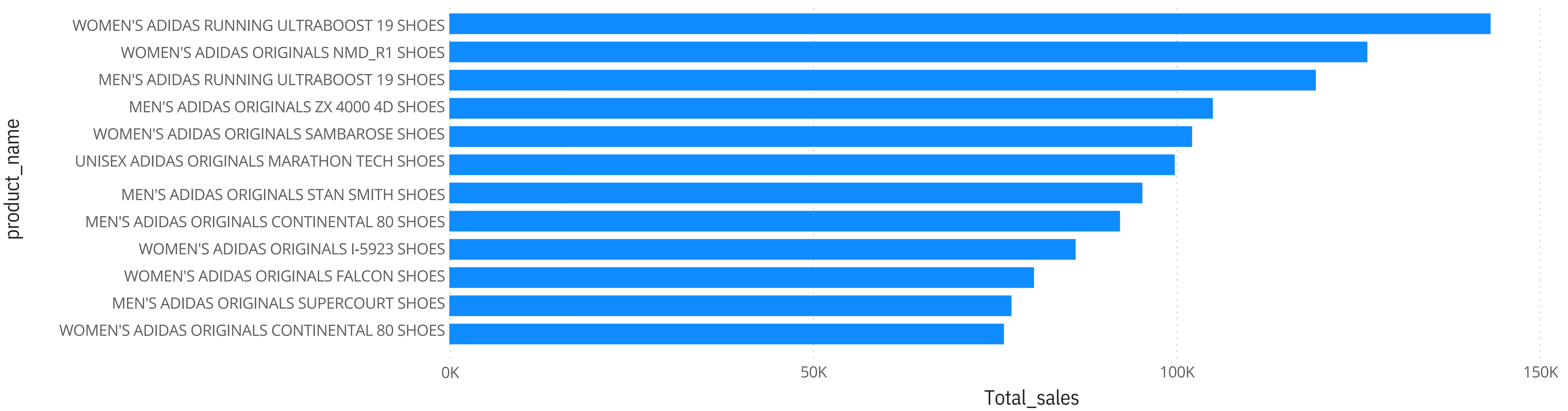
SENTIMENT ANALYSIS USING VADER (NLTK)

Footwear Sales & Sentiment Dashboard

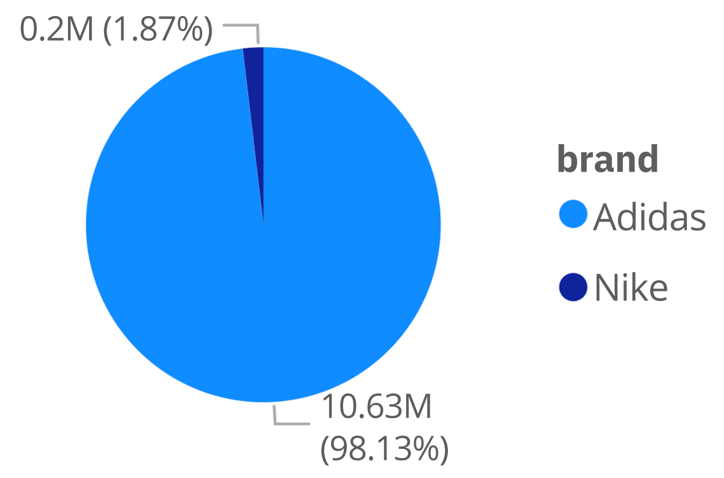
10.83M
Total_sales

last_order_date
1/01/2018
12/04/2020

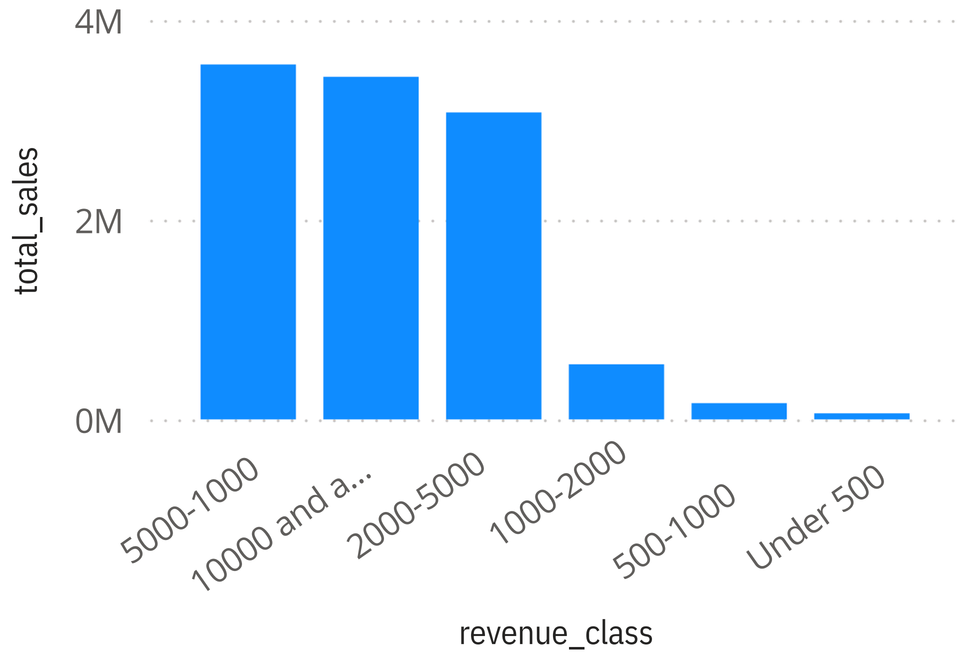
Sales per product



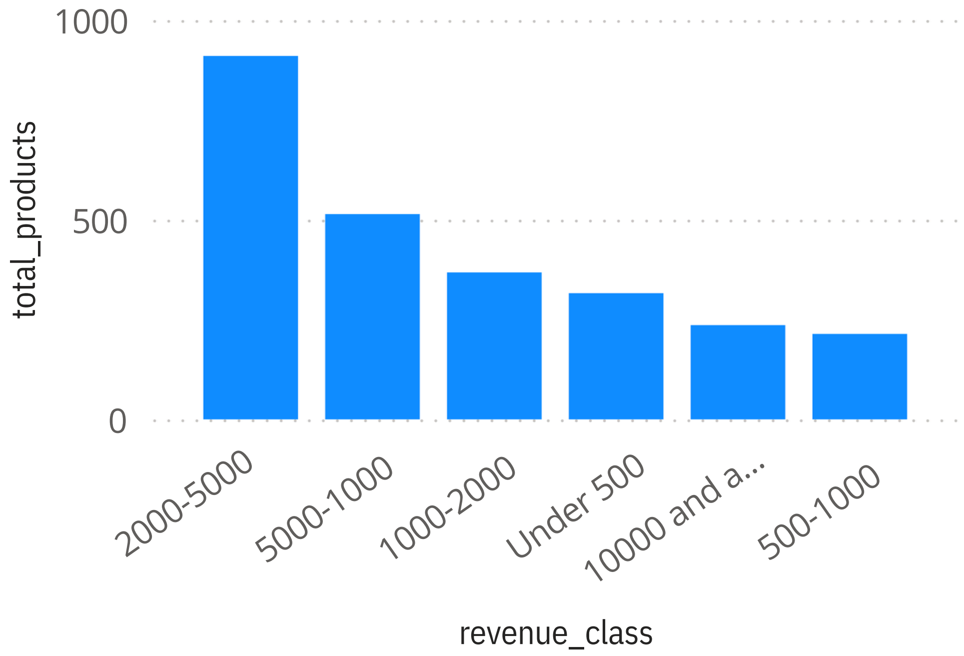
Total sales by Brand



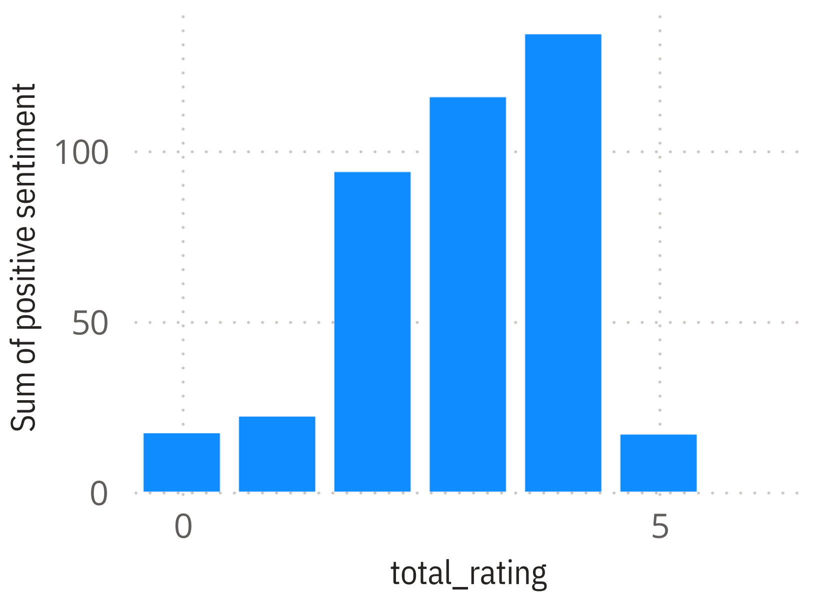
Total sales by Spending Class



Number of Products Sold by Spending Class



Positive Sentiment Analysis of the Description vs Review Score



TECK STACK

SQL Server
ETL (Custom SQL workflows)
Python - VADER NLP
Power BI
Draw.io
GitHub

SPECIAL THANKS:

Special thanks to Baraa Khatib Salkini for his excellent educational content and guidance on implementing SQL-based data warehouse architectures.